

# Final Report — North vs South India Regional Classification

Full audit trail · Zomato Delivery dataset · Team Hmmmmmmmmmm · 18:30 deliverable

**One-paragraph summary.** We classified 41,944 Zomato orders as **South** (16,397, 39.1%) or **North** (25,547, 60.9%) using city codes embedded in Delivery\_person\_ID. All 22 city codes verified within 45 km of the expected centroid. We then trained three increasingly strict models: **(A) GEO** using raw coordinates (AUC 1.000, calibration); **(B') BEHAVIOUR-only** after removing a temporal data-collection leak (AUC 0.508, essentially random); and **(C) STRUCTURAL**, adding only the haversine distance between pickup and drop (AUC 0.996/0.911 StratKFold / GroupKFold). The finding has three layers: **behaviour** is uniform across India; **geometric structure** (the spatial distribution of restaurants and customers within cities) differs significantly between regions; **geography** (absolute coordinates) is perfectly separable by construction.

## 1. Methodology

### 1.1 Region labelling

Each rider ID follows the pattern {CITY}RES##DEL##. A regex extracts the prefix; a 22-row dictionary maps prefix → city → state → region.

Per the team brief: **South** = Kerala, Tamil Nadu, Andhra Pradesh, Telangana, Karnataka. **North** = every other state (including Maharashtra, Gujarat, Goa, West Bengal, MP, Rajasthan, Punjab, UP, Uttarakhand, Jharkhand).

### 1.2 Coordinate sanity check (city centroid vs median lat/lon of city's orders)

| Code   | City       | State          | Region | n     | med-lat | med-lon | km off | OK |
|--------|------------|----------------|--------|-------|---------|---------|--------|----|
| AURG   | Aurangabad | Maharashtra    | North  | 555   | 19.876  | 75.346  | 0.8    | ✓  |
| VAD    | Vadodara   | Gujarat        | North  | 1,576 | 22.311  | 73.168  | 1.3    | ✓  |
| JAP    | Jaipur     | Rajasthan      | North  | 3,443 | 26.905  | 75.795  | 1.7    | ✓  |
| COIMB  | Coimbatore | Tamil Nadu     | South  | 3,169 | 11.009  | 76.976  | 2.2    | ✓  |
| RANCHI | Ranchi     | Jharkhand      | North  | 2,563 | 23.359  | 85.325  | 2.6    | ✓  |
| ALH    | Allahabad  | Uttar Pradesh  | North  | 568   | 25.452  | 81.835  | 2.9    | ✓  |
| KNP    | Kanpur     | Uttar Pradesh  | North  | 676   | 26.475  | 80.321  | 2.9    | ✓  |
| KOL    | Kolkata    | West Bengal    | North  | 700   | 22.539  | 88.365  | 3.5    | ✓  |
| MYS    | Mysore     | Karnataka      | South  | 3,011 | 12.323  | 76.628  | 3.5    | ✓  |
| PUNE   | Pune       | Maharashtra    | North  | 3,132 | 18.546  | 73.843  | 3.5    | ✓  |
| LUDH   | Ludhiana   | Punjab         | North  | 687   | 30.893  | 75.822  | 3.7    | ✓  |
| AGR    | Agra       | Uttar Pradesh  | North  | 550   | 27.162  | 78.043  | 3.8    | ✓  |
| DEH    | Dehradun   | Uttarakhand    | North  | 492   | 30.343  | 78.061  | 3.9    | ✓  |
| INDO   | Indore     | Madhya Pradesh | North  | 3,158 | 22.745  | 75.888  | 4.0    | ✓  |
| SUR    | Surat      | Gujarat        | North  | 3,187 | 21.171  | 72.789  | 4.2    | ✓  |

| Code | City      | State          | Region | n     | med-lat | med-lon | km off | OK |
|------|-----------|----------------|--------|-------|---------|---------|--------|----|
| BHP  | Bhopal    | Madhya Pradesh | North  | 478   | 23.232  | 77.43   | 4.3    | ✓  |
| BANG | Bangalore | Karnataka      | South  | 3,193 | 12.97   | 77.64   | 5.5    | ✓  |
| CHEN | Chennai   | Tamil Nadu     | South  | 3,144 | 13.03   | 80.242  | 6.3    | ✓  |
| MUM  | Mumbai    | Maharashtra    | North  | 3,173 | 19.122  | 72.835  | 6.7    | ✓  |
| KOC  | Kochi     | Kerala         | South  | 701   | 9.986   | 76.297  | 6.9    | ✓  |
| HYD  | Hyderabad | Telangana      | South  | 3,179 | 17.431  | 78.4    | 10.6   | ✓  |
| GOA  | Goa       | Goa            | North  | 609   | 15.517  | 73.767  | 44.9   | ✓  |

All 22 cities passed the 250-km threshold; worst offender Goa at 45 km from centroid (the rider-ID prefix is correct).

## 2. Univariate EDA — the apparent flat finding

Eleven structured statistical tests on individual features. Mann-Whitney U for numeric, Chi-square for categorical, KS for distribution-shape on order\_hour. ALL eleven returned  $p > 0.05$ .

| #             | Feature                         | Test           | Statistic   | p        | Verdict       |
|---------------|---------------------------------|----------------|-------------|----------|---------------|
| 01_target     | Delivery time differs by region | Mann-Whitney U | U=207941110 | 2.13e-01 | North > South |
| 02_weather    | Weather distribution differs    | Chi-square     | chi2=7.7    | 2.64e-01 | Same          |
| 03_traffic    | Traffic density differs         | Chi-square     | chi2=1.0    | 7.95e-01 | Same          |
| 04_multi      | Multi-delivery norms differ     | Mann-Whitney U | U=199693348 | 3.90e-01 | North > South |
| 05_age        | Rider age skew                  | Mann-Whitney U | U=192135843 | 6.51e-01 | North > South |
| 06_rating     | Rider rating skew               | Mann-Whitney U | U=191711132 | 9.11e-01 | North > South |
| 07_vehicle    | Vehicle preferences differ      | Chi-square     | chi2=3.9    | 2.76e-01 | Same          |
| 08_order_type | Order-type mix differs          | Chi-square     | chi2=0.7    | 8.78e-01 | Same          |
| 09_festival   | Festival incidence differs      | Chi-square     | chi2=2.3    | 1.32e-01 | Same          |
| 10_hour       | Time-of-day rhythm              | KS 2-sample    | D=0.006     | 8.30e-01 | Same          |
| 11_city_tier  | City-tier mix differs           | Chi-square     | chi2=3.0    | 2.29e-01 | Same          |

**Layer 1 conclusion.** No single behavioural feature distinguishes North from South orders at conventional significance. Delivery time, traffic density, weather, vehicle type, multi-deliveries, rider age and rating, festival incidence, time-of-day, city-tier mix — eleven tests, zero hits. The naive read of this is 'regions are identical.' That read is incomplete, as section 4 shows.

## 3. Two-model baseline — geography vs behaviour

### 3.1 Model A — raw coordinates

Features: Restaurant lat/lon, Delivery lat/lon, distance\_km. 5-fold StratifiedKfold, seed 42, LightGBM with class\_weight='balanced'. **Mean AUC = 1.0000**, F1 = 1.000, balanced accuracy = 1.000. Per-fold AUCs were all 1.0000.

Verdict: labels are clean. This model is for calibration only — coords trivially separate regions because we defined the regions by state which is a function of coords.

### 3.2 Model B — behaviour, naive (has temporal leak)

Features: 12 numeric + 6 categorical, no coordinates. **Mean AUC = 0.5879**. SHAP on this model showed month dominating (importance 0.45, 10× the next feature).

### 3.3 Temporal leak diagnosed

Date range is identical for both regions (Feb 11 – Apr 6 2022). But within-region month %s differ:

| Region | February | March | April |
|--------|----------|-------|-------|
| North  | 20.8%    | 66.0% | 13.2% |
| South  | 4.3%     | 79.9% | 15.8% |

Feb is 4.8× more represented in North than South orders — a data-collection-window artifact, not seasonality. month is therefore a region-proxy, not a behavioural signal, and must be dropped.

### 3.4 Model B' — behaviour, leak-free

Drop month and day\_of\_week; same model, same split. **Mean AUC = 0.5076**, F1 = 0.433. Per-fold: 0.5141, 0.5153, 0.5005, 0.5005, 0.5079. Two folds at exactly 0.500 — random.

**Layer 1 conclusion (rigorous).** Even with a tuned tree ensemble and class-weight correction, behaviour-only features cannot tell North from South at better than chance once the data-collection leak is removed. Eleven univariate tests + a behaviour-only model both say the same thing: **operational behaviour is uniform across India.**

## 4. Deep dive — six stress tests

Before signing off on 'no behavioural signal', we ran six independent stress tests in case interaction effects or distribution-shape differences were hiding the pattern.

### 4.1 KS distribution-shape tests

Mann-Whitney compares medians; KS compares the entire distribution. Even when medians are equal, KS picks up shape differences.

| Feature             | KS D   | p        | Significant? |
|---------------------|--------|----------|--------------|
| distance_km         | 0.0879 | 6.70e-67 | YES          |
| is_late_night       | 0.0089 | 4.12e-01 | no           |
| Time_taken (min)    | 0.0080 | 5.39e-01 | no           |
| is_weekend          | 0.0079 | 5.58e-01 | no           |
| prep_time_min       | 0.0073 | 7.97e-01 | no           |
| Delivery_person_Age | 0.0067 | 7.77e-01 | no           |

distance\_km is the only column with a significant shape difference ( $D = 0.088$ ,  $p \approx 6.7 \times 10^{-67}$ ). Every other feature has near-identical distributions.

### 4.2 Interaction ablation under GroupKFold

To make sure we are not over-claiming via per-city memorization, we use **GroupKFold by city** — train on some cities, test on others. We then add one engineered feature at a time and watch the AUC.

| Feature added (on top of base behaviour) | GroupKFold AUC | Lift    |
|------------------------------------------|----------------|---------|
| distance per age                         | 0.8780         | +0.3722 |
| traffic x hour                           | 0.5076         | +0.0019 |
| vehicle x multi                          | 0.5036         | -0.0022 |
| weather x traffic                        | 0.5023         | -0.0035 |
| rating x traffic                         | 0.5008         | -0.0050 |
| (baseline, no extras)                    | 0.5058         | 0.0000  |

Four of five engineered interactions add zero lift. The one with measurable lift (distance\_per\_age) was simply distance\_km in disguise — the denominator (age) was incidental.

### 4.3 Behaviour + distance\_km under GroupKFold (per-fold)

| Fold | Held-out cities            | Structural AUC |
|------|----------------------------|----------------|
| 1    | AGR, ALH, JAP, LUDH, MYS   | 0.9981         |
| 2    | BANG, PUNE, RANCHI         | 0.9979         |
| 3    | BHP, CHEN, SUR, VAD        | 1.0000         |
| 4    | DEH, HYD, INDO, KNP, KOC   | 0.6422         |
| 5    | AURG, COIMB, GOA, KOL, MUM | 0.9165         |

Mean GroupKFold AUC = 0.9109. Fold 4 dropped to 0.642 — that fold held both Kochi (Kerala) and Hyderabad (Telangana). When all our 'South' training data came from Karnataka and Tamil Nadu, generalisation to Kerala+Telangana was imperfect. This is itself an insight: **South India is internally heterogeneous in its geometric layout.**

#### 4.4 Per-South-state classifier (is South internally homogeneous?)

A 4-way classifier on the South subset (states: Karnataka, Kerala, Tamil Nadu, Telangana) using only behaviour features reaches accuracy **0.377**, against a random baseline of 0.250. 38% vs 25% is a 13 percentage-point lift — modest but real. Within-South heterogeneity confirms the fold-4 result above.

#### 4.5 CatBoost cross-check

To rule out a LightGBM-specific artefact, we retrained Model B' as a CatBoost classifier on the same split. Mean AUC = **0.5070** — within 0.001 of the LightGBM value. The 'no behavioural signal' result is not framework-dependent.

## 5. Layer 2 — the structural model

Adding only one feature to the behaviour-only model — the haversine distance between pickup and drop — pushes the GroupKFold AUC from 0.5076 (random) to **0.9109**. StratKFold AUC = 0.9956. F1 = 0.952, balanced accuracy = 0.964.

### Why this is structural, not geographic:

- distance\_km is an **order-level scalar**, not a location. It is the great-circle distance between the restaurant and the customer for a single order. Two orders in different cities can have identical distance\_km values.
- Its distribution differs between regions (KS D = 0.088,  $p \approx 6.7 \times 10^{-10}$ ) even when medians match — implying South cities have a different *shape* to their delivery-distance histogram than North cities.
- GroupKFold (train on some cities, test on others) preserves most of the lift — proving the pattern transfers to unseen cities, not just rote per-city memorisation.
- SHAP shows distance\_km dominating the structural model (4.87 vs 0.15 for the next feature) — the signal is concentrated and identifiable, not spread across a noisy ensemble.

**What this means operationally.** South Indian cities have a measurably different spatial distribution of restaurants and customers than North Indian cities. The behaviour of riders, the weather, the traffic, the order types — all uniform. But the geometry of the city itself is different enough that a model can pick up on it from delivery-distance histograms alone. This is a finding about **urban form**, not about Zomato's operation.

## 6. The three-layer conclusion

| Layer        | What we measured                                                                             | Honest AUC | Read                                  |
|--------------|----------------------------------------------------------------------------------------------|------------|---------------------------------------|
| 1. Behaviour | Rider, customer, weather, traffic, time, festival, vehicle — everything that can be measured | 1.000      | Uniform across India                  |
| 2. Structure | Haversine distance distribution per order                                                    | 0.911      | Differs significantly                 |
| 3. Geography | Raw lat/lon coordinates                                                                      | 1.000      | Perfectly separable (by construction) |

**Headline finding.** Zomato's operations are remarkably uniform across India — North and South orders look identical in every behavioural dimension we measured. The only honest non-geographic difference between regions is the **geometric layout** of cities, reflected in the distribution of pickup-to-drop distances. South cities are *structurally* different but *operationally* the same as North cities.

## 7. Limitations & honest caveats

- Data window is only 8 weeks (Feb 11 – Apr 6 2022). A longer window could reveal seasonal patterns (Onam, Diwali, Pongal) that this window misses.
- Only 22 cities. A wider sample would tell us whether the 'structural difference' generalises to tier-2/3 South Indian cities not in our data.
- South Indian sub-regions are unbalanced — Kerala (KOC) has only 701 orders vs Karnataka (BANG+MYS) at 6,200. The Fold-4 anomaly reflects this.
- Time-of-day is on order placement, not delivery time-of-day — late deliveries could drift across hour boundaries.
- The model is trained with class\_weight='balanced'; raw posterior probabilities are NOT well-calibrated for absolute interpretation, only for ranking.
- We do not have a held-out temporal test split — generalisation to a future week is unverified.

## 8. Artefacts on GitHub

| Path                                       | Description                                                  |
|--------------------------------------------|--------------------------------------------------------------|
| src/regional.py                            | City-code → region mapping + lat/lon fallback + sanity check |
| notebooks/_run_regional.py                 | 11 EDA tests + Models A and B (raw, leak-detection)          |
| notebooks/_run_regional_deep.py            | Six deep-check stress tests                                  |
| notebooks/_build_regional_deck.py          | Pitch-deck generator (live JSON)                             |
| notebooks/_build_final_report.py           | This document                                                |
| reports/regional_findings.json             | Every test, every metric, every confusion matrix             |
| reports/regional_city_check.csv            | 22-city centroid validation                                  |
| reports/figures/fig_reg_*.png              | 11 EDA + 4 model plots + ablation comparison                 |
| reports/pitch_regional_630pm.{pptx,pdf}    | 10-slide pitch deck                                          |
| models/region_classifier_v1.txt            | Model B' booster (behaviour-only, leak-free)                 |
| models/region_classifier_v2_structural.txt | Model C booster (behaviour + distance_km)                    |

**Repo:** [github.com/TheClazer/Zomato-delivery-estimation](https://github.com/TheClazer/Zomato-delivery-estimation) · **Team:** Hmmmmmmmmmm (Suchit SM · Rayyan Shaikh · Ranadeep M) · **Generated:** from live regional\_findings.json by \_build\_final\_report.py